MA.912.AR.3.5

Benchmark

MA.912.AR.3.5	**Given the x-intercepts and another point on the graph of a quadratic function, write the equation for the function.**

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.1.3, MA.912.AR.1.7
- MA.912.F.1.6

Terms from the K-12 Glossary

- Quadratic Function
- x-intercept
- y-intercept

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.2

Next Benchmarks

- MA.912.AR.3.4

Purpose and Instructional Strategies

In grade 8, students determined the slope in a linear relationship when given two points on the line. In Algebra I, students write the equation for a quadratic function when given the x-intercepts and another point on the graph. In later courses, students will write the equation for a quadratic function when given a vertex and another point on the graph.

- Instruction includes making connections to various forms of quadratic equations to show their equivalency. Students should understand and interpret when one form might be more useful than another depending on the context.
  - Standard Form
    Can be described by the equation $y = ax^2 + bx + c$, where a, b and c are any rational number. This form can be useful when identifying the y-intercept.
  - Factored form
    Can be described by the equation $y = a(x - r_1)(x - r_2)$, where r_1 and r_2 are real numbers and the roots, or x-intercepts. This form can be useful when identifying the x-intercepts, or roots.
  - Vertex form
    Can be described by the equation $y = a(x - h)^2 + k$, where the point (h, k) is the vertex. This form can be useful when identifying the vertex, which is the turning point of the function where the axis of symmetry intersects.
- Instruction includes comparing and contrasting between a linear function of the form $y = a(x - h) + k$ and a quadratic function of the form $y = a(x - h)^2 + k$. This will also extend to an absolute value function of the form
  $y = a\,|x - h| + k$. *(MTR.5.1)*
- Instruction includes the use of x-y notation and function notation.
- Instruction includes the use of graphing technology.

Common Misconceptions or Errors

- Similar to their work with vertex form, students may have trouble with using the correct signs for r_1 and r_2 in factored form. In these cases, show students a corresponding graph of their developed factored form functions to help them see the need for opposite signs in their factors.
- Student might multiply incorrectly when squaring a binomial. For example, $(x - h)^2$ is incorrectly distributed as $x^2 - h^2$ instead of $x^2 - 2xh + h^2$.
- When rewriting their developed factored form functions into standard form, students may incorrectly multiply a by both factors. One remedy for this is to direct students to multiply their binomials first and then multiply the product by a.

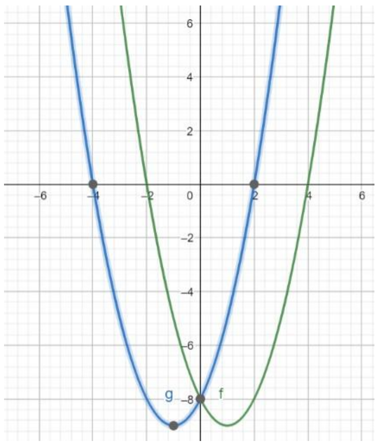
Strategies to Support Tiered Instruction

- Teacher provides instruction to show how different methods can be used to multiply binomials after substituting the roots into factored form. Then, ask students to recall methods to multiply binomials. For example, direct students to multiply their binomials first and then multiply the product by a.
- Instruction includes converting a quadratic from vertex form to standard form.
- Instruct students to write out $(x - h)^2$ as $(x - h)(x - h)$. Remind students that different methods can be used to multiply binomials.
- Instruction includes an opportunity to discuss multiplicity of roots when converting from factored form to standard form. The teacher can use graphing software to build the connection between the standard form of a quadratic function with a perfect square trinomial and the resulting parabola with only one root. Students see that the root should be used for both r_1 and r_2 in factored form and confirm by converting their resulting function back to standard form.
    - For example, students could see a problem that only presents one root and express confusion on how it applies to the factored form. These cases provide an opportunity to discuss multiplicity of roots.
- Teacher models how to solve for a by substituting the values of x and y from a point on the parabola.
- For students who need support evaluating functions, review the steps for evaluating a function given an input value by co-creating an anchor chart.

Given Parabola: $y = 6x^2 - 5x + 11$	
Input: -4	Output
$y = 6(-4)^2 - 5(-4) + 11$	$y = 127$

-

- Teacher provides opportunities to use an online graphing tool to graph an equation with the x-intercepts to help students who may have used the wrong signs in their factors.
    - For example, if the roots of a quadratic function are 2 and −4, the teacher can provide the graphs of the two functions (as shown below) to determine which equation corresponds to the roots.



Blue graph: $y = (x - 2)(x + 4)$
Green graph: $y = (x + 2)(x - 4)$

Instructional Tasks

Instructional Task 1 (MTR.7.1)

A city fountain shoots jets of water that pass back and forth through a marble wall in its center. One jet of water begins 12 feet away from the wall and passes through a hole in the wall that is 12 feet high before landing 5 feet away on the other side. Write a quadratic function that represents the path the jet of water takes.

Instructional Task 2 (MTR.3.1, MTR.6.1, MTR.7.1)

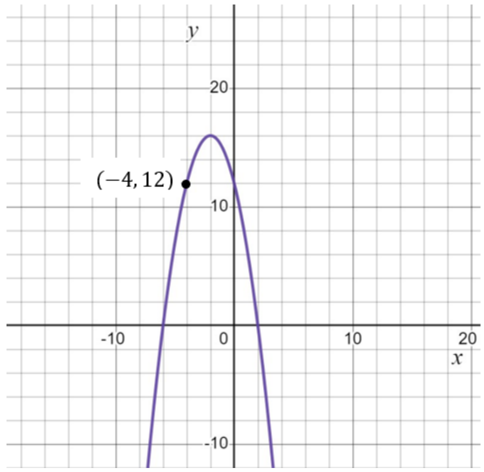
As part of a math project, you are analyzing the profit (P) of a smartphone manufacturing company based on the number of units sold (x). The company's profit can be modeled by a quadratic function. During your research, you discover that the company experiences a break-even point when no profit is made, and this occurs when no smartphones are sold. Additionally, when the company sells 1 smartphone, it makes a profit of $6.

Write the quadratic equation that represents the company's profit, using the given information about the break-even point and the profit when selling one smartphone.

Instructional Items

Instructional Item 1

Write a quadratic function to represent the graph below.



Instructional Item 2

Write a quadratic function to represent a parabola with roots of 5 and -7 that passes through the point $(-3, -128)$.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*